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Paleomagnetism of the Neoproterozoic Chuar Group, Grand Canyon Supergroup, Arizona: implications for Laurentia's Neoproterozoic APWP and Rodinia break-up

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Abstract

Selected parts of the Galeros and Kwagunt Formations, of the Neoproterozoic Chuar Group (ca. 800–740 Ma) exposed in the eastern Grand Canyon, Arizona, yield paleomagnetic data that are consistent with a primary origin. Both formations are dominated by dark mudrock with multiple, hematite-cemented fine sandstone and mudstone horizons that yield magnetizations of west declination and shallow inclination (or antipodes). The grand mean magnetization for the older Galeros Formation, determined by giving unit weight to each of 16 sites, is Dec = 265.5°, Inc = 5.1°, $\alpha 95 = 8.5°$, $\kappa = 19.8$ and yields a paleopole of 2°S, 163°E (dp = 4.3; dm = 8.5). The younger Kwagunt Formation provides a grand mean magnetization of Dec = 280.8°, Inc = 30.6°, $\alpha 95 = 8.4°$, $\kappa = 38.6$ (three sites of normal polarity and six sites of reverse polarity), determined by giving unit weight to each of the nine sites, and yields a paleopole of 18°N, 166°E (dp = 5.2; dm = 9.4). The NRM of about a third of the collection, mainly from coarser-grained hematitic sandstones, is dominated by a north-directed, moderate positive inclination magnetization with laboratory unblocking temperatures up to 680 °C and is interpreted as a recent overprint. We argue that the east–west and shallow magnetization in the Chuar Group strata, although similar to early Paleozoic magnetizations for North America, is primary because both groups of data pass fold tests and because Chuar strata lie in angular unconformity below the flat-lying Middle Cambrian Tapeats sandstone. Furthermore, we infer that the Chuar Group paleopoles, together with other North American Neoproterozoic results, are fully consistent with a tight counterclockwise loop in the south Pacific in the apparent polar wander path for North America during the Neoproterozoic. This new path represents a drift history that positions the central part of the Rodinia supercontinent at or near the equator at ~800 Ma and slightly younger time. Penecontemporaneous with Chuar deposition, extension began along the Cordilleran margin of Laurentia. This large-scale extensional event culminated in the rift-to-drift transition of Laurentia away from East Gondwana or Asian (South China, Siberia) blocks by ca. 750 Ma.
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1. Introduction

Well-dated Neoproterozoic paleomagnetic poles for Laurentia are relatively sparse and, consequently, considerable uncertainty exists about the paleogeog-

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raphy of Laurentia for this time period (Roy, 1983; Park, 1994; Buchan et al., 2001; Meert and Powell, 2001). Conversely, some intervals during the Mesoproterozoic for Laurentia (i.e., the Logan-Loop) provide the most complete Precambrian apparent polar wander paths (APWP) of any of the major continents. The Logan-Loop is largely derived from rocks of the Keweenawan rift system of northeastern Laurentia (1.1–1.0 Ga) and is defined by over 60 poles having, overall, well-determined stratigraphy, structure and geochronology (Halls and Pesonen, 1982). On the other hand, for the Neoproterozoic, and more specifically for the time interval between 1.0 and 0.75 Ga, most paleopoles have been derived from high-grade metamorphic rocks from the Grenville Province of northeastern Laurentia, which have been either chemically and or thermally remagnetized (McWilliams and Dunlop, 1978; Hyodo and Dunlop, 1993; Costanzo and Dunlop, 1998; Warnock et al., 2000). Although Laurentia's APWP has no substantial spatial discontinuities between the Keweenawan and Grenville pole sequences, the direction and sense of younging of the Grenville aged part of the APWP remains contested (McWilliams and Dunlop, 1978; Berger et al., 1979; Roy, 1983; Hyodo and Dunlop, 1993; Weil et al., 1998).

Global paleogeography for this time period (1.1–0.75 Ga), and in particular Laurentia's positions, are especially important in the context of the hypothesized Rodinia supercontinent and its history of amalgamation and subsequent fragmentation (Hoffman, 1991; Dalziel, 1991, 1997; Weil et al., 1998; Meert and Powell, 2001; Hartz and Torsvik, 2002; Torsvik, 2003). Rodinia is hypothesized as a long-lived Proterozoic supercontinent formed from the amalgamation of all of the major continental blocks during the Grenville orogeny (∼1.10 Ga). However, both the ultimate configuration and lifecycle of Rodinia are still poorly understood (Buchan et al., 2001; Meert and Powell, 2001; Torsvik, 2003). Of particular interest and contention are the timing of Rodinia's breakup and the paleogeography of the dispersing continents during breakup.

Surrounded by rift margins that developed in the Neoproterozoic (Bond et al., 1984), Laurentia is thought to have made up the core of Rodinia during the supercontinent's lifecycle. Proposals for possible connecting pieces include Australia, East Antarctica, South China, and Siberia for Laurentia's Cordillera and Arctic margins, and Baltica, Amazonia, Rio de la Plata, Congo, and Kalahari for Laurentia's Appalachian margin (Piper, 1982; Moores, 1991; Hoffman, 1991; Li et al., 1996; Dalziel, 1997; Weil et al., 1998; Torsvik et al., 2001a, 2001b; Karlstrom et al., 2001). Although there is compelling evidence for the approximate timing of Laurentia's 'break-out' from Rodinia (Hoffman, 1991), and the subsequent opening of the Paleo-Pacific and Iapetus Oceans, large discrepancies still exist between available paleomagnetic data and the geologic records of continent dispersal (Lindsay et al., 1987; Bond and Kominz, 1984; Powell et al., 1993, 1994; Meert and Powell, 2001).

One of the most widely accepted adjoining blocks for Laurentia's Cordilleran margin is Australia (Bell and Jefferson, 1987; Moores, 1991; Karlstrom et al., 2001). Stratigraphic similarities between Laurentia and Australia, and subsidence curves of the Laurentia margin suggest a Vendian age of separation (Bond and Kominz, 1984) between East Gondwana and Laurentia. However, paleomagnetic results from Laurentia and East Gondwana indicate an earlier rift-to-drift age ranging between ∼800 and 725 Ma (Powell et al., 1993; Wingate et al., 2002). This 200 Ma difference in the timing of rift initiation is substantial and has important ramifications for global tectonics. It is now widely accepted that formation of the Gondwana supercontinent—amalgamation of East and West Gondwana tectonic elements—was completed by ca. 560 Ma along the Mozambique belt of eastern Africa during the East African Orogen (Stern, 1994). Recent geochronologic and paleomagnetic data support a two-stage model for the collision of East and West Gondwana, with final amalgamation culminating at ca. 550 Ma (Meert et al., 1995; Meert and Van der Voo, 1996). Clearly, if these interpretations are correct, the East Gondwana blocks must have rifted away from the Cordilleran margin of Laurentia considerably earlier than predicted by the thermal subsidence models for the western margin of Laurentia (Bond et al., 1984). This dichotomy emphasizes the need for additional, well-dated paleomagnetic poles from both East Gondwana and Laurentia for the time interval between 900 and 700 Ma to better define the timing and paleogeography of Rodinia's breakup.

A better understanding of Neoproterozoic paleogeography is also needed due to the recent renewed

interest in Harland's (1964) hypothesis that the Earth experienced a long-lived widespread ice age in the Neoproterozoic. Kirschvink (1992) proposed, on the basis of paleomagnetic data interpreted to indicate that many of these global glacial deposits formed at equatorial latitudes, a "snowball" Earth hypothesis, during which the planet is covered in ice, weathering reactions basically cease, and carbon dioxide concentrations

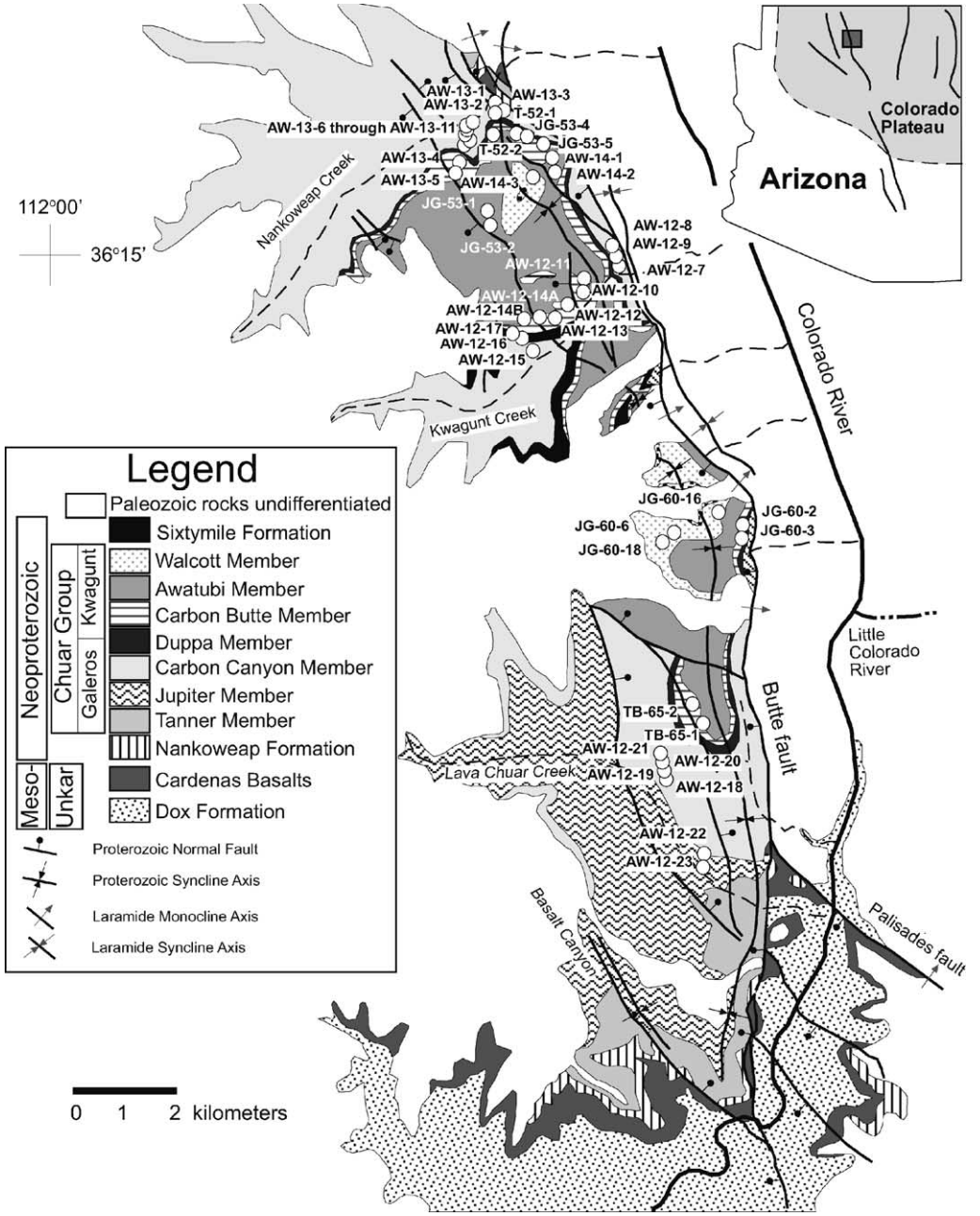


Fig. 1. Simplified geologic map of sampled Proterozoic rocks exposed in the Grand Canyon, Arizona. Modified from Timmons et al. (2001).

gradually increase in the atmosphere, ultimately creating a global, CO_2-induced catastrophic warming. Hoffman et al. (1998), Hoffman (1999) and Hoffman and Schrag (2002) assimilated stratigraphic and stable isotope data that strongly support such a hypothesis and in fact argue that the development of a snowball Earth and consequent rapid warming may have taken place up to four times in the Neoproterozoic. With a growing body of information supporting periods of exceedingly unusual climate behavior for Earth in the Neoproterozoic, additional, robust paleomagnetic data for this time period have become even more important. Some initial paleomagnetic data from Neoproterozoic rocks from North America (e.g., Elston, 1989a, 1989b; Elston, 1993; Park and Aitken, 1986) are consistent with the possibility of North America lying at or near the equator during this time period, but there are issues of concern that bear on the overall reliability of many of the data, especially the data from the American Southwest. These include inadequate documentation of demagnetization results, inaccuracies in the age of the rocks sampled, and the possibility that the magnetizations are secondary, as the poles derived from many Neoproterozoic rocks have positions similar to those from rocks of Cambro-Ordovician age.

We report here new paleomagnetic results from the Chuar Group, which forms the upper part of the Grand Canyon Supergroup (GCSG), northern Arizona (Fig. 1). Exposed in a 6 km × 24 km area in the eastern Grand Canyon, the sedimentary sequence is one of the thickest (about 1600 m thick) and best-preserved Neoproterozoic successions in North America. Because depositional ages for these rocks span between ~800 and 740 Ma (Karlstrom et al., 2001; Timmons et al., 2001), these rocks were studied with the hope that they could provide a temporally well-defined set of paleopoles that better elucidates Laurentia's Neoproterozoic APWP, and thereby provides an improved definition of the general paleogeography of the Neoproterozoic.

2. Geologic setting

The Chuar Group comprises a sequence of shallow marine and tidal-flat deposits. The sequence is dominated by mudrock (>85%) with subordinate meter-scale beds of dolomite and sandstone. The Chuar Group has been subdivided into the Galeros and Kwagunt Formations, which are further subdivided into seven members (Fig. 2) (Ford and Breed, 1973). Internal deformation of the Chuar Group is minor (i.e. broad folds and minor faults) and interpreted to be related to large-scale extension prior to 740 Ma (Timmons et al., 2001). Subsequent Laramide deformation produced high-angle reverse faults and locally juxtaposed Chuar Group and Paleozoic strata.

The age of the Chuar Group is bracketed between 1.09 Ga (underlying Cardenas Basalts) and about 0.74 Ga (ash at the top of the Chuar Group (Dehler et al., 1999)). The basal age comes from a Rb-Sr whole-rock date of 1070 ± 70 Ma and a $^{40}Ar/^{39}Ar$ age of 1091 ± 5 Ma derived from the underlying Cardenas Basalts and its feeder dikes (Elston and McKee, 1982; Timmons et al., 2001; Weil et al., 2003). The upper age limit of 742 ± 6 Ma is based on a U–Pb zircon date of an ash bed from the Walcott member (Karlstrom et al., 2000). Fossil assemblages including *Chuaria*, *Baicalia-Boxonia*, *Stratifera-Inzeria* and vase-shaped microfossils found throughout the Chuar Group succession corroborate a Neoproterozoic age (Vidal and Ford, 1985; Porter and Knoll, 2000). Preliminary paleomagnetic work in the Chuar Group is also consistent with a Neoproterozoic age for the Chuar succession (Elston 1989a,b, 1993; also summarized in Elston et al., 2002).

Environmental interpretation of Chuar Group facies has been the subject of substantial debate, mainly due to the lack of megafossils and bioturbation (Ford and Breed, 1973; Vidal and Knoll, 1983; Reynolds et al., 1988; Cook, 1991; Dehler et al., 2001a). Recently, Dehler et al. (2001a) used microfossils, sedimentary structures, facies associations, and geochemical trends to discount previously held non-marine interpretations for Chuar deposition (Ford and Breed, 1973; Reynolds et al., 1988; Cook, 1991). All Chuar facies indicate a marine setting with no unequivocal evidence suggesting terrestrial depositional environments (Dehler et al., 2001a). Ripple-laminated and cross-bedded sandstone facies exhibit epsilon cross-beds, bi-directional trough cross-beds, interference ripples, convolute bedding, load casts, and mudcracked mud drapes that overlie symmetrical ripples, which collectively suggest a marine tidal environment (Dehler et al., 2001a). The existence of marine fossils and high local pyrite

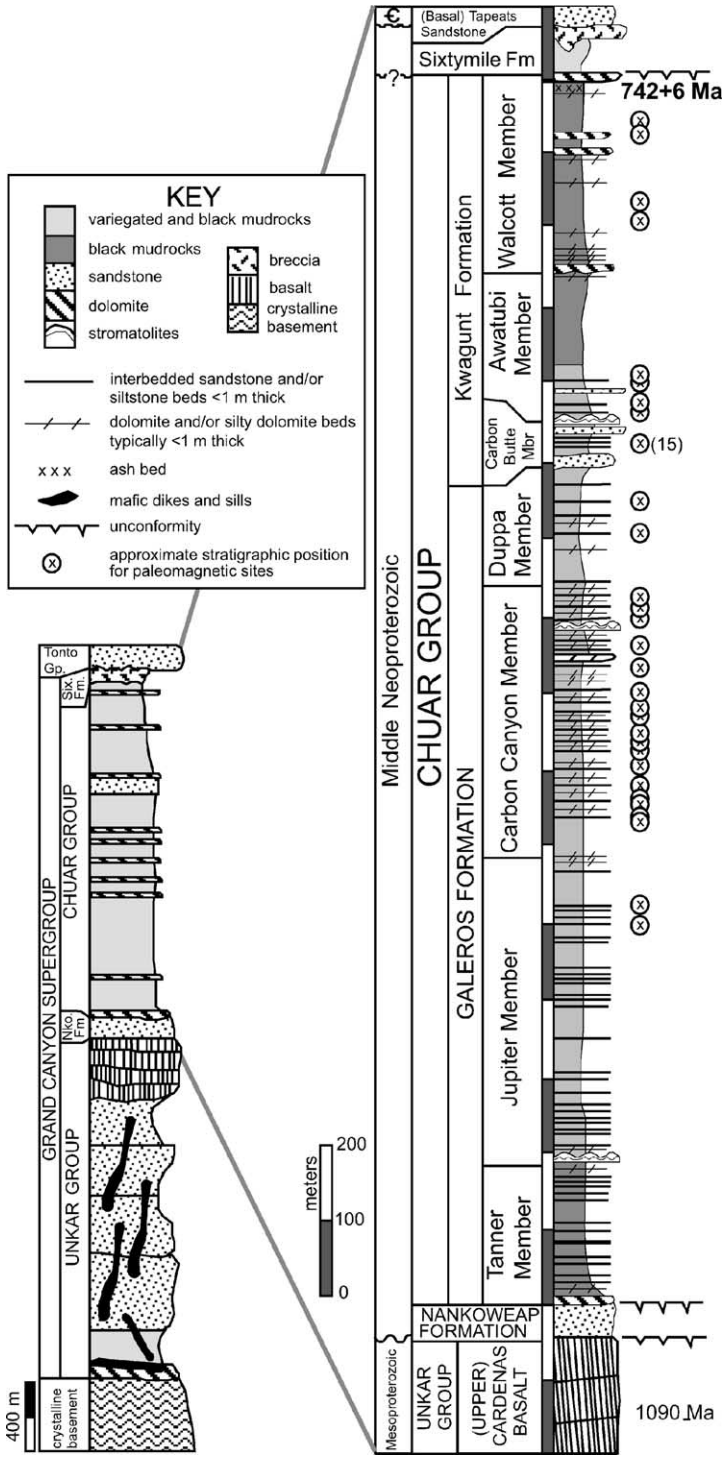


Fig. 2. Simplified stratigraphic section of the Grand Canyon Supergroup, Arizona. Column modified from Dehler et al. (2001a,b).

content within the mudrocks also indicates a marine environment (Porter and Knoll, 2000).

It has recently been proposed that the Chuar Group was part of a large intra-continental seaway that formed during initial phases of extension and rifting along the Cordilleran margin (Dehler et al., 2001b). In their model, Dehler et al. (2001b) hypothesize that the Chuar Group was deposited penecontemporaneously with rifting in a large sea, which they call the CHUMP (Chuar, Uinta Mountain and Pahrump groups) seaway, additional remnants of which are represented by parts of the Uinta Mountain Group of northeastern Utah and the Pahrump Group of the Death Valley area, California.

Structural interpretations of the Chuar Group indicate deposition concurrent with east–west extension along north–south trending structures, including the well-exposed Butte fault (Timmons et al., 2001). The main structural feature in the Chuar Group is the Chuar syncline just west of, and parallel to, the Butte fault (Fig. 1). Timmons et al. (2001) documented several features from the Chuar syncline that they interpreted to be consistent with a growth fold origin for the structure. These features include tightening of the syncline with depth, intraformational faults, soft-sediment deformation, stratigraphic thickening within the hinge zone of the fold, and abrupt stratigraphic and thickness variations across the Butte fault. The trends of the Chuar syncline, as well as the orientation of syn-depositional normal faults like the Butte fault, suggest that it formed subparallel to the ancient Cordilleran rift margin (Timmons et al., 2001).

Stratigraphic and structural interpretations of the Chuar Group indicate the succession was deposited in an intracratonic basin created as a consequence of extension, possibly related to the breakup of Rodinia (Sears, 1990; Timmons et al., 2001). Detailed descriptions of Chuar stratigraphy, structure and paleontology are found in Dehler et al. (2001a), Timmons et al. (2001) and Porter and Knoll (2000), respectively.

3. Methods

With numerous red sandstone, siltstone and mudstone beds, the Chuar Group is ideal for a comprehensive paleomagnetic study. Forty-four sites were collected in the Kwagunt and Galeros Formations from a total of six different members. Sampling was completed in the summers of 1999–2001. Site locations were distributed within the side canyons off of the main Colorado River corridor. Due to National Park Service restrictions, oriented hand samples were collected in the field using a geologic hammer and chisel. Seven to nine hand samples were collected per site, with each site typically consisting of multiple beds. Severe outcrop weathering and highly faulted areas were avoided where possible. Bedding and sample orientations were measured in the field using a magnetic compass. Sampling concentrated on hematite-rich detrital rocks, with grain sizes typically ranging from medium sand to mud.

On average, eight standard 2.54 cm diameter paleomagnetic specimens from each site were progressively demagnetized in an Analytical Service Co. (ASC) thermal demagnetizer and measured in a three-axes Cryogenic 2G Magnetometer in the field-free room at the University of Michigan's paleomagnetic laboratory. Samples from several sites were demagnetized in the field-free room at the University of New Mexico's paleomagnetic laboratory, using either a Shaw Magnetic Measurements or a Schonstedt TSD-1 furnace. Thermal heating was carried out in 100 °C increments to 300 °C, with smaller temperature steps to a maximum temperature of 680 °C. Chemical demagnetization (Henry, 1979) was carried out in the University of New Mexico Laboratory. Remanence directions were calculated from principal component analysis (Kirschvink, 1980) of linear vectors picked from orthogonal projection demagnetization plots (Zijderveld, 1967). Site means were calculated by averaging the sample directions, using the method of Fisher (1953). Fold tests were performed where possible to characterize the relative age of magnetization acquisition.

4. Results

Two statistically unique paleomagnetic poles were derived from Chuar Group strata, one from the Kwagunt and one from the Galeros Formations. Demagnetization behavior of Chuar Group rocks is variable but overall the rocks yield uniform dual polarity characteristic remanent magnetizations (ChRMs) of high coercivity and high, distributed laboratory unblocking temperature (Figs. 3 and 4).

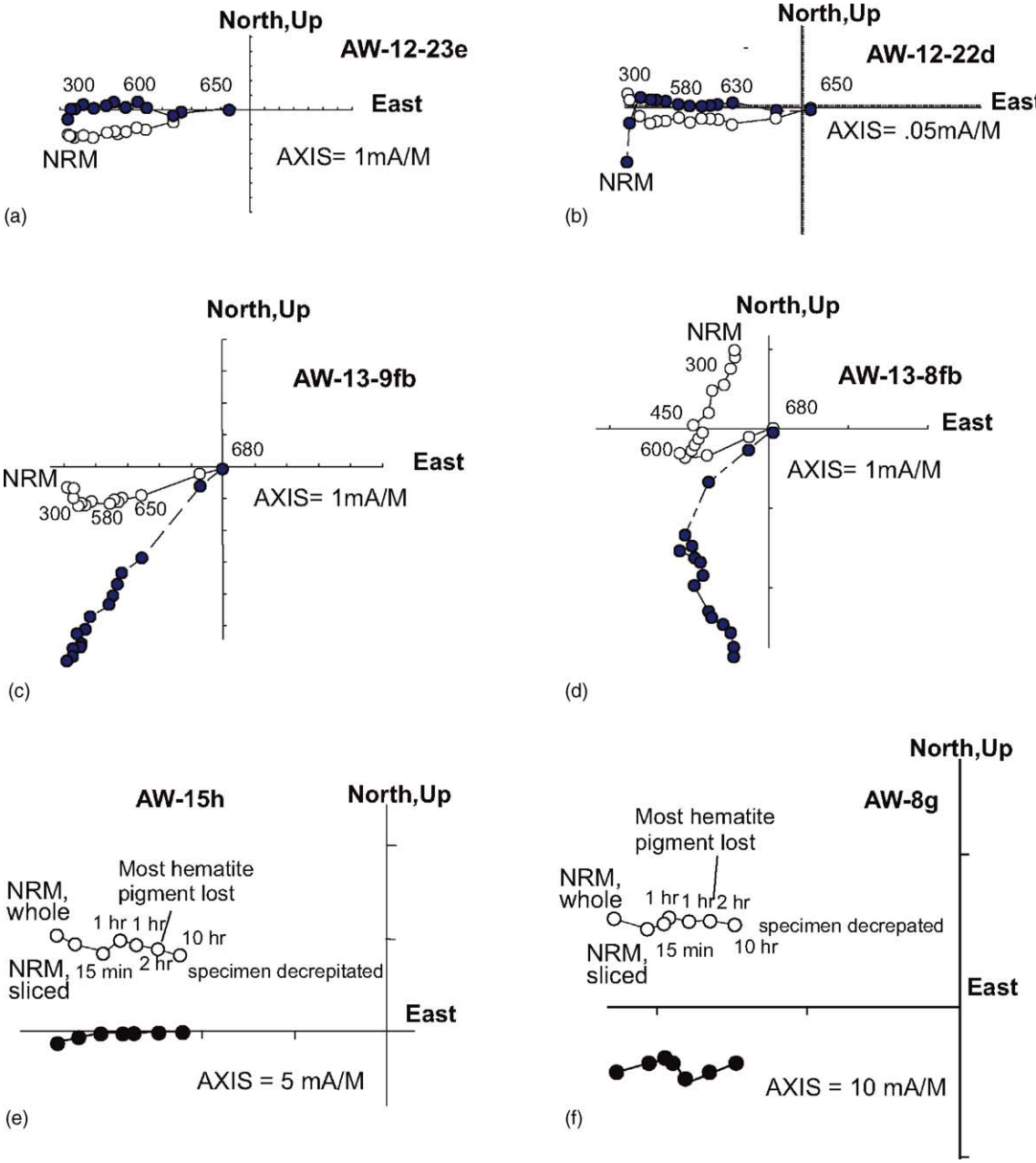


Fig. 3. (a–d) Orthogonal demagnetization plots, in geographic coordinates, of representative samples from the Galeros Formation. Filled (open) symbols indicate projections onto the horizontal (vertical) plane. (e and f) Orthogonal demagnetization plots, in geographic coordinates of representative chemical demagnetization (Henry, 1979) response of samples from the upper part of the Galeros Formation. Chemical demagnetization times indicate the duration of the individual leaching step, not the cumulative duration of leaching. After the 10 h step, most hematite-cemented Chuar rocks are partly to wholly decrepitated. Note that most of the hematite pigment is removed by leaching after a few hours, and specimens reach a yellowish gray (5 Y 7/2) to grayish yellow (5 Y 8/4) color.

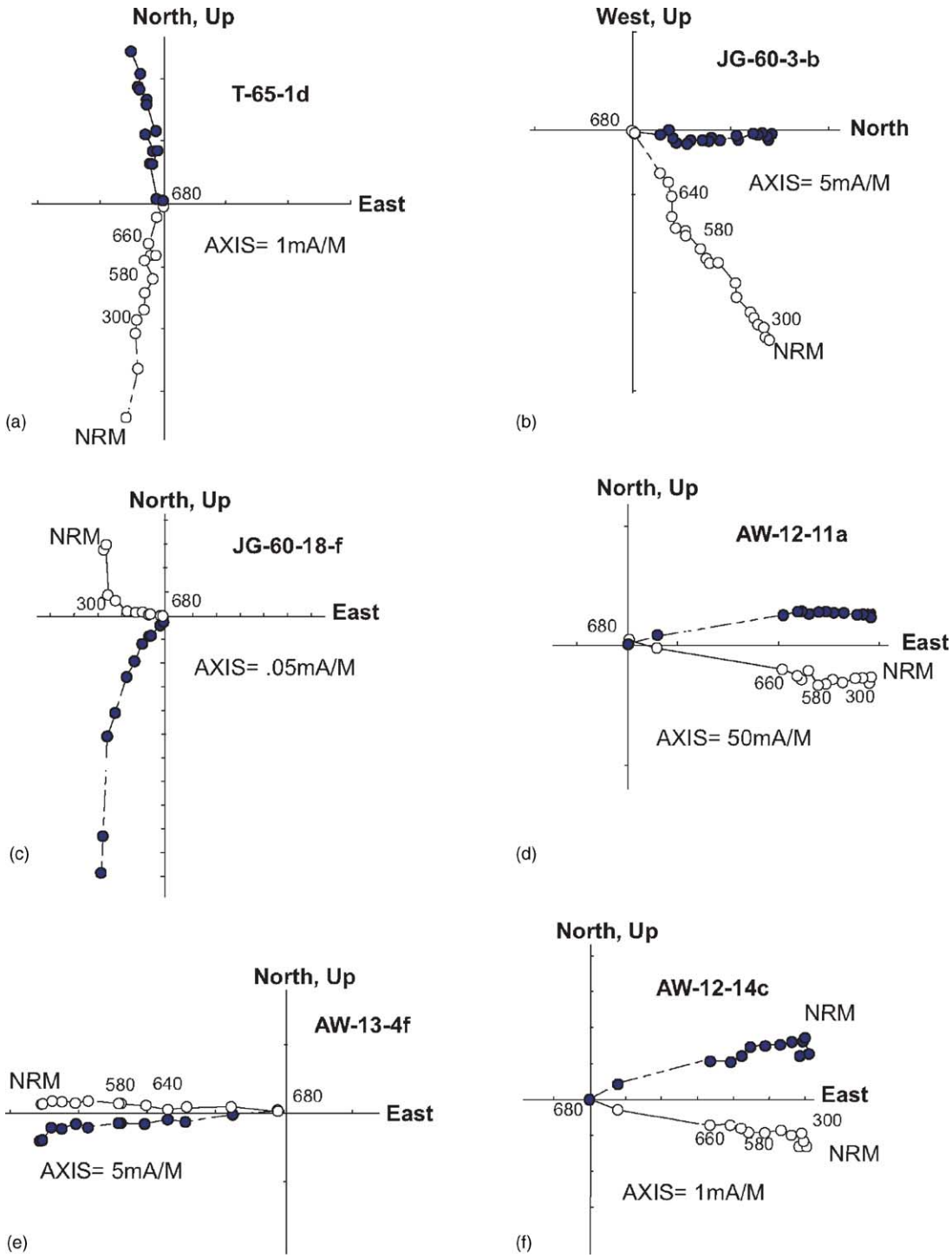


Fig. 4. Orthogonal demagnetization plots, in geographic coordinates, of representative samples from the Kwagunt Formation. Filled (open) symbols indicate projections onto the horizontal (vertical) plane.

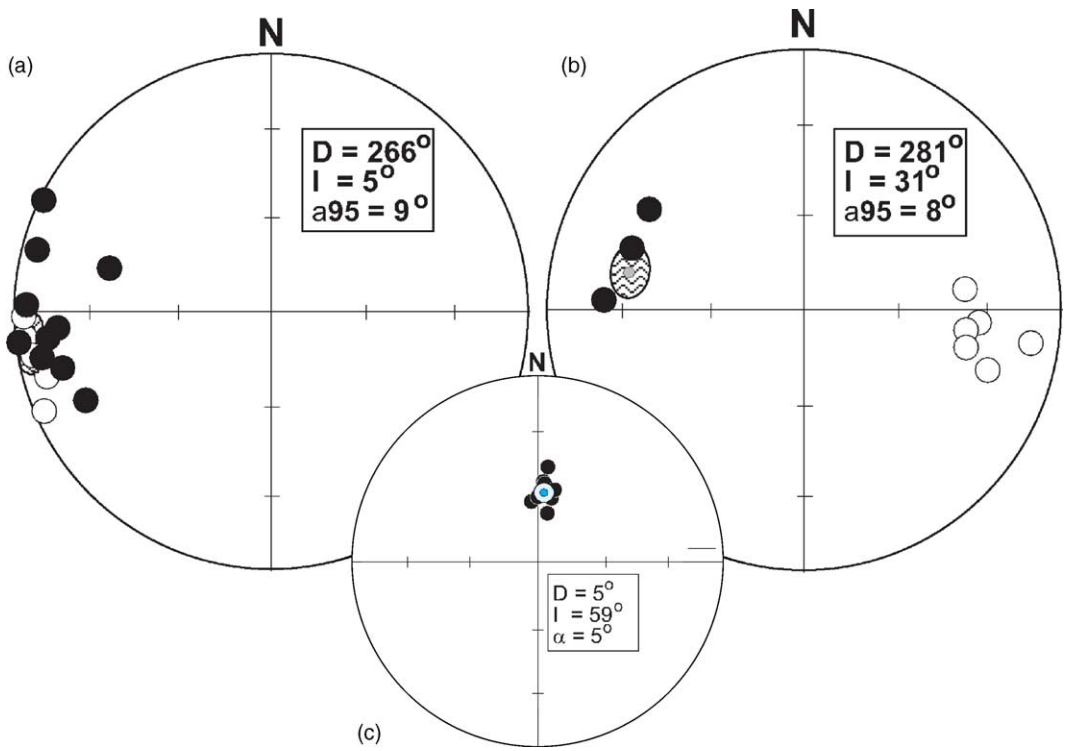


Fig. 5. Equal area projections showing structurally corrected paleomagnetic site mean directions (filled symbols: lower hemisphere projection) for the (a) Galeros Formation and the (b) Kwagunt Formation. (c) PDF overprint sites ($N = 9$ sites) from the Carbon Butte Member of the Kwagunt Formation.

4.1. Galeros Formation

Intensities of natural remanent magnetizations (NRM) had values between 0.1 and 20.0 mA/m. Most of the Galeros samples had discrete unblocking temperatures between 630 and 680 °C, indicating hematite to be the main remanence carrier (Fig. 3). Galeros sites were dominated by magnetizations of west declinations and moderate to shallow inclinations (Fig. 5a). However, some individual samples carried an opposite polarity magnetization of east and shallow direction.

In stepwise demagnetization (Fig. 3), a north-directed, present-day field viscous overprint was usually removed by 300 °C. After removal of any PDF or recent overprint, most Galeros samples showed the isolation of a well-defined high unblocking temperature (∼680 °C) component. The 16 sites that carried a stable ChRM are well-grouped and have an acceptable within-site dispersion (Table 1). The remaining five sites either did not carry a stable remanence direction, or yielded magnetizations of unacceptably high dispersion at the site level.

The sites in Galeros Formation strata were distributed throughout both limbs of the Chuar syncline, whose north-trending fold axis provided an ideal fold test given the general orientation of the ChRM in these rocks. The in-situ site mean magnetizations have inclinations that range from +60 to −30°, and upon tilt correction have a much more clustered range. Incremental unfolding yields a maximum kappa value ($\kappa = 20$) at 100% unfolding (Fig. 6a). This result is interpreted as a pre-folding magnetization. To test the robustness of the incremental fold test a parametric bootstrap fold test was applied (Tauxe, 1998), which gives a similar result of maximum clustering at ∼100% unfolding (Fig. 6b).

The mean paleopole of the Galeros Formation, determined by giving unit weight to each of the 16 sites is 2°S, 163°E (dp = 4.3; dm = 8.5) (Table 2).

Table 1
Paleomagnetic site information and statistical parameters for the Chuar Group

Site number	Strike	Dip	N/N_0	In situ site mean				Structurally corrected site means	
				Dec	Inc	$\alpha 95$	κ	Dec	Inc
Kwagunt Formation									
Walcott member									
AW-14-3[a]	261	2	0/7	–	–	–	–	–	–
JG-60-16[a]	81	6	0/8	–	–	–	–	–	–
JG-60-18	200	24	7/7	290	54	18	27	303	29
JG-60-6	223	21	7/7	299	50	13	28	290	30
Awatubi member									
AW-14-1[a]	159	35	0/8	–	–	–	–	–	–
AW-14-2[a]	175	29	0/7	–	–	–	–	–	–
JG-53-1[a]	144	23	0/7	–	–	–	–	–	–
JG-53-2	139	30	6/7	107	−31	11	76	98	−13
Carbon Butte member									
T-52-1[b]	148	22	8/8	5	64	8	56		
T-65-1[b]	340	18	5/8	4	54	11	50		
T-65-2[b]	340	18	6/8	5	53	7	107		
JG-60-2[b]	195	30	6/8	0	61	6	122		
JG-60-3[b]	185	46	6/8	5	55	8	151		
JG-53-4[b]	155	29	6/8	13	57	10	51		
JG-53-5[b]	155	29	9/9	345	63	10	30		
AW-13-4	5	20	8/8	272	3	9	39	272	23
AW-13-5[b]	45	22	8/8	6	51	7.8	61		
AW-12-10	313	14	6/8	88	−25	12	65	94	−34
AW-12-11	332	21	6/8	103	−13	16	258	108	−28
AW-12-12[b]	334	22	8/8	11	68	9	61		
AW-12-13	16	9	7/8	85	−28	21	20	83	−36
AW-12-14A	341	21	6/8	92	−20	12	60	97	−39
AW-12-14B	341	21	8/8	97	−19	12	39	103	−37
Galeros Formation									
Duppa member									
AW-12-16[a]	303	22	0/7	–	–	–	–	–	–
AW-12-17[a]	318	24	0/7	–	–	–	–	–	–
Carbon Canyon member									
AW-12-7[a]	168	53	0/8	–	–	–	–	–	–
AW-12-8	162	43	6/8	260	60	13	27	256	17
AW-12-9	160	60	7/8	256	48	9	90	254	−11
AW-12-15	303	12	7/8	277	31	15	24	285	35
AW-12-18[a]	333	35	0/8	–	–	–	–	–	–
AW-12-19	310	39	7/7	299	−8	16	16	296	1
AW-12-20	290	16	6/7	284	5	7	53	285	7
AW-12-21[a]	255	15	0/8	–	–	–	–	–	–
AW-13-1	110	25	7/7	264	6	11	31	264	−5
AW-13-2	105	20	7/7	264	18	15	16	259	10
AW-13-3	85	20	7/8	254	26	14	21	245	22
AW-13-6	35	30	7/8	241	−20	12	31	247	−5
AW-13-7	37	29	6/7	260	−19	17	22	263	2
AW-13-8	25	13	7/7	267	−2	14	24	267	10
AW-13-9	30	30	6/7	252	−29	15	27	259	−7

Table 1 (*Continued*)

Site number	Strike	Dip	N/N_0	In situ site mean				Structurally corrected site means	
				Dec	Inc	$\alpha 95$	κ	Dec	Inc
AW-13-10	25	23	7/7	264	−6	14	25	263	13
AW-13-11	30	30	7/8	263	−30	19	13	269	−5
Jupiter member									
AW-12-22	287	17	8/8	268	−11	16	20	266	−5
AW-12-23	75	1	8/8	270	3	6	112	272	5

N/N_0 is ratio of samples used in the calculation of the site mean direction to the total number of samples collected and measured; strike/dip, dip direction is 90° clockwise from strike; Dec and Inc are declination and inclination of the site mean directions; $\alpha 95$ is semi angle of the cone of confidence about the site mean direction; κ is precision parameter (Fisher, 1953).

[a] Designates those sites that had no interpretable results.

[b] Represents those sites in which all samples carry a stable PDF component upto 680 °C.

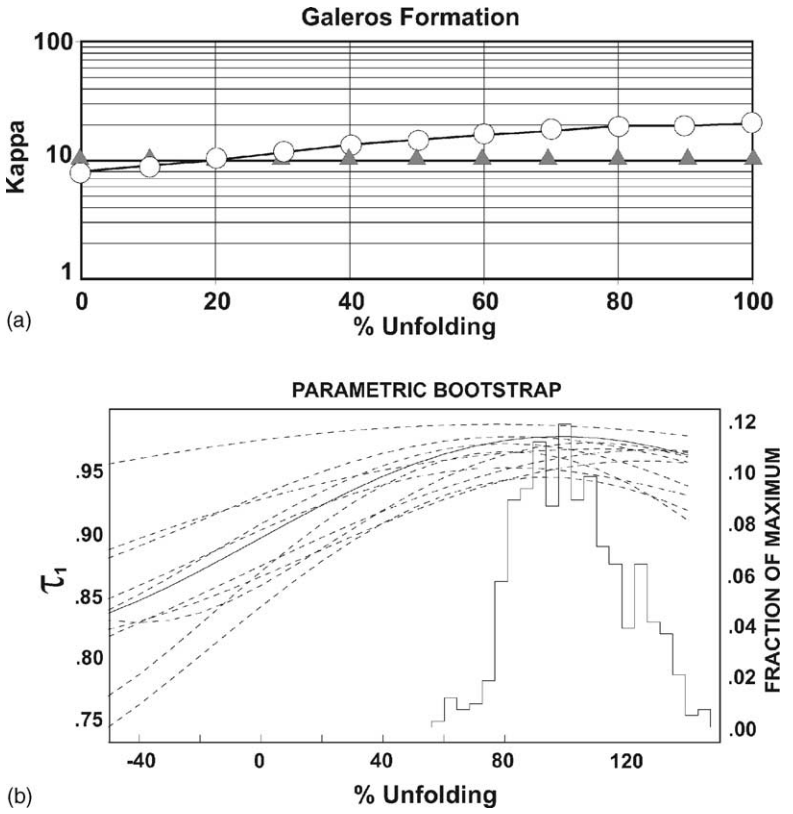


Fig. 6. (a) Incremental fold test for the Galeros Formation, plotting the kappa parameter (open circles) vs. percent unfolding (McElhinny, 1964). Maximum kappa ($\kappa = 23$) reached at 100% unfolding. (b) Incremental parametric bootstrap fold test for the Galeros Formation (Tauxe, 1998). The τ_1 parameter represents the largest eigenvalue of the orientation matrices from representative para-data sets. Histogram represents 500 maxima of τ_1 with their respective fraction of maximum given by the right-hand side ordinate axis. Calculated 95% confidence interval is between 77 and 132% unfolding.

Table 2
Paleomagnetic site information for the Chuar Group, Grand Canyon Supergroup, Arizona

Pole location	Approximate age (Ma)	N/N_0 sites	In situ				Structurally corrected				Plong.	PLat.	dp	dm
			Dec	Inc	α95	κ	Dec	Inc	α95	κ				
Chuar Group														
Kwagunt Formation	740	9/23	278.0	27.3	12.2	18.6	280.8	30.6	8.4	38.6	166	18.2	5.2	9.4
Galeros Formation	800	16/21	265.7	4.0	14.0	7.9	265.5	5.1	8.5	19.8	163	−2.1	4.3	8.5

N/N_0 is ratio of sites used in the calculation of the grand mean direction to the total number of sites collected and measured; Dec and Inc are declination and inclination of the mean paleomagnetic directions; α95 is semi-angle of the cone of confidence about the mean direction; Plong. and Plat. are the latitude and longitude of the mean paleomagnetic pole; dp/dm are the semi-axes of the cone of 95% confidence about the pole; the value of the paleomagnetic pole in the table is based on the tilt corrected site mean directions.

4.2. Kwagunt Formation

Intensities of natural remanent magnetizations (NRM) for Kwagunt Formation samples were similar to those of the Galeros Formation samples with values ranging from 0.1 to 30.0 mA/m. Most of the Kwagunt samples had discrete unblocking temperatures between 650 and 680 °C, also indicating hematite

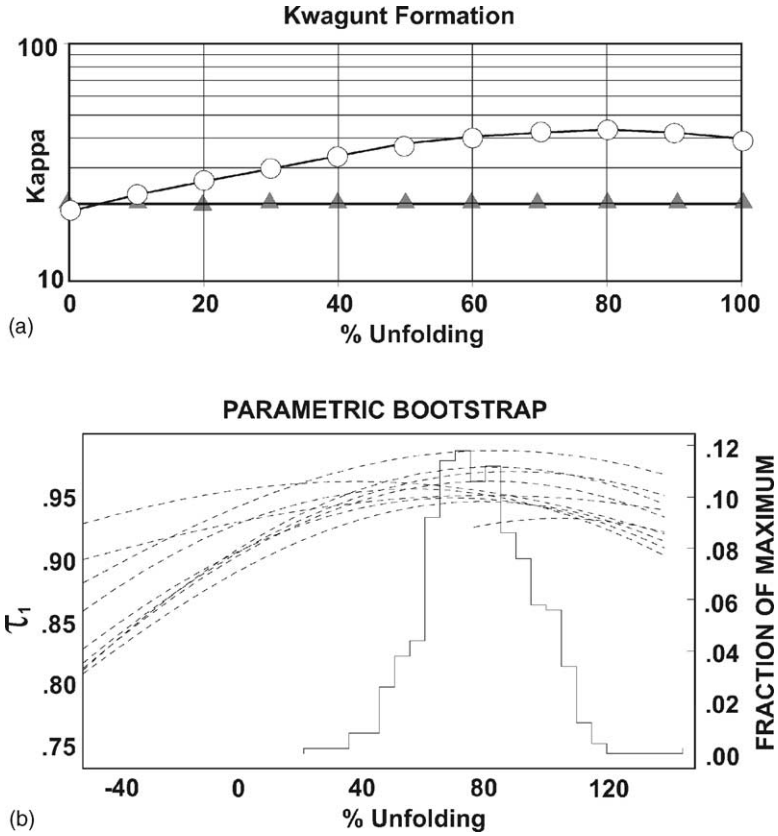


Fig. 7. (a) Incremental fold test for the Kwagunt Formation, plotting the kappa parameter (open circles) vs. percent unfolding (McElhinny, 1964). Maximum kappa ($\kappa = 44$) reached at 80% unfolding. (b) Incremental parametric bootstrap fold test for the Kwagunt Formation (Tauxe, 1998). The τ_1 parameter represents the largest eigenvalue of the orientation matrices from representative para-data sets. Histogram represents 500 maxima of τ_1 with their respective fraction of maximum given by the right-hand side ordinate axis. Calculated 95% confidence interval is between 52 and 107% unfolding.

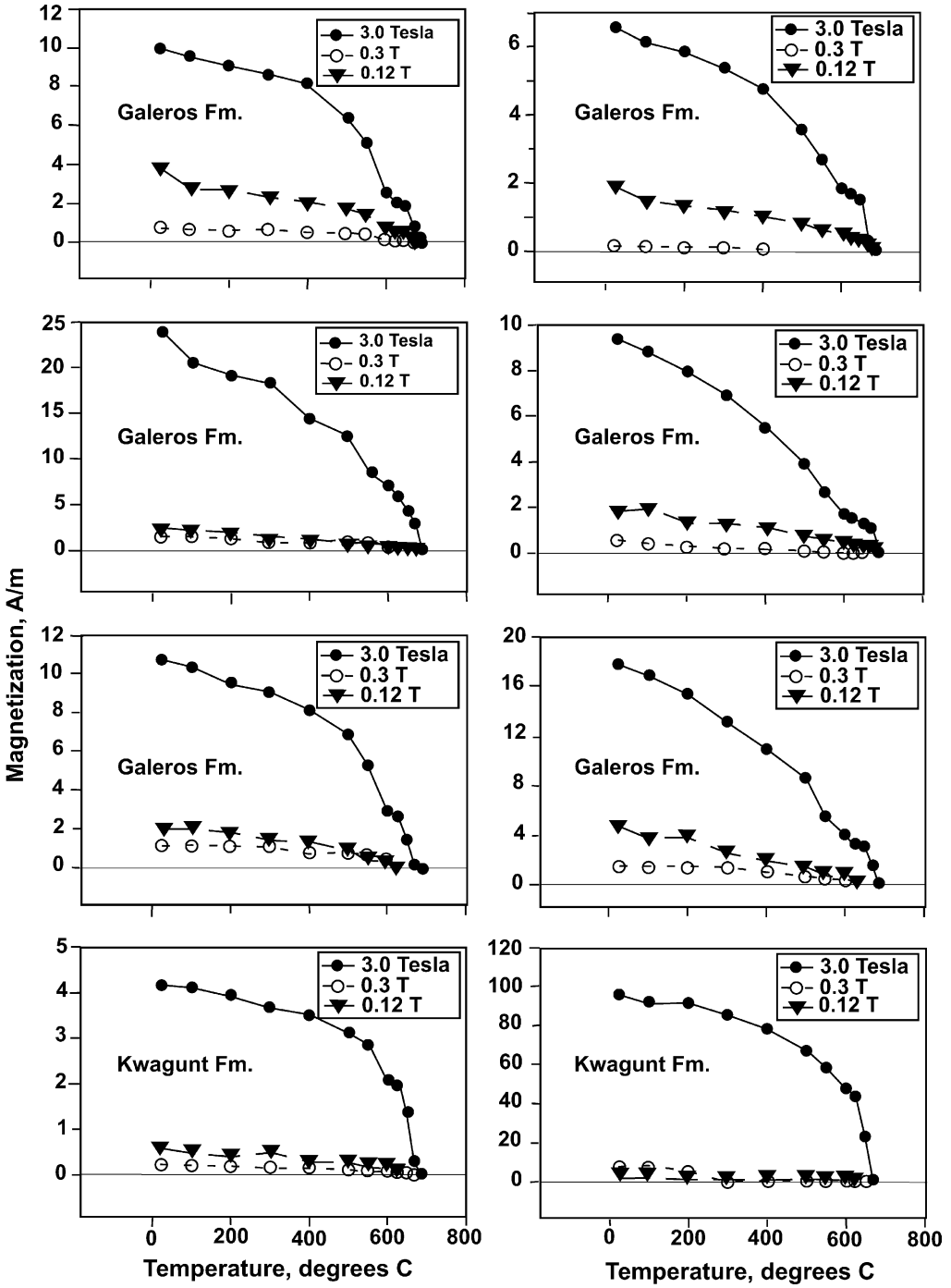


Fig. 8. Intensity vs. temperature plots during three-component IRM thermal demagnetization, acquired along the specimen *Z*-axis (3.0 T), *Y* axis (0.3 T) and *X*-axis (0.12 T), following the method of Lowrie (1990).

to be the main remanence carrier (Fig. 4). Kwagunt sites had west-declination and moderate positive inclination directions as well as east-directed and intermediate negative directions (Fig. 5b). A McFadden and Lowes (1981) reversal test was done on all nine Kwagunt sites indicating that the opposite polarity populations overlap at the 95% level of confidence and that the two means cannot be distinguished. The presence of a positive reversal test suggests that that progressive demagnetization of Kwagunt samples was successful and that any secondary component has been removed from analysis.

In stepwise demagnetization, a north-directed present-day field viscous overprint was usually removed by 300 °C. However, in two-thirds of the Carbon Butte Member sites, the unblocking temperature spectrum for the PDF overprint ranged up to 680 °C (Fig. 4a and b), and no recognizable ancient magnetization could be isolated (Fig. 5c). After removal of any recent overprint in those samples that did carry an ancient ChRM, a well-defined univectorial high temperature component was retrieved up to laboratory unblocking temperatures of about 680 °C (Fig. 4c–f). The nine Kwagunt sites that carried a stable ChRM are well-grouped and associated with low dispersion at the site-level (Table 1). The remaining 14 sites either were characterized by unacceptably high dispersion of high laboratory unblocking temperature magnetizations (five sites), a strong viscous PDF overprint or the absence of a recognizable ancient ChRM (nine sites).

As with the Galeros Formation, the sites from the Kwagunt Formation were distributed throughout both limbs of the Chuar syncline, affording an ideal fold test. The results of a McElhinny (1964) fold test are given in Fig. 7a. Incremental unfolding yields a maximum kappa value ($\kappa = 44$) at 80% unfolding, and we interpret the result to indicate a pre-folding magnetization given the syn-depositional nature of Chuar deformation. To test the robustness of the incremental fold test a parametric bootstrap fold test was also applied (Tauxe, 1998), which gives a similar result of maximum clustering at ∼80% unfolding (Fig. 7b). The clear evidence for syn-depositional folding in the Chuar (Timmons et al., 2001) combined with the pre to early-folding magnetization interpreted for both the Galeros and Kwagunt Formations is clear support for a primary magnetization acquired at the time of, or soon after, deposition.

The mean paleopole of the Kwagunt Formation, determined by giving unit weight to each of the nine sites is 18°N, 166°E (dp = 5.2; dm = 9.4) (Table 2).

4.3. Rock magnetism

Following the method of Lowrie (1990), we subjected samples from several representative sites that yielded well-defined, well-grouped magnetizations to progressive thermal demagnetization of isothermal remanent magnetization (IRM) applied using fields of 3.0, 0.3, and 0.12 Tesla (T) along three orthogonal axes (Fig. 8). All samples reveal that the principal magnetic phase present is hematite. In all samples, the magnetization acquired in the intermediate field value is very low. The presence of minor quantities of a lower coercivity magnetic phase is indicated by the fact that the magnetization acquired in a 0.12 T field is up to about 20% of that acquired in a field of 3.0 T. Notably, the hematite-cemented sandstones and siltstones that have yielded well-grouped, early-acquired magnetizations do not contain a substantial amount of magnetite, which we would infer would have been of detrital origin. Chemical demagnetization (Henry, 1979) of selected, representative materials reveals a less than 20% reduction in the NRM intensity during leaching in a total of 4 h (Fig. 3e and f). Notably, over this time period most of the specimens lost essentially all of their red coloration, which we interpret to indicate that most pigmentary hematite was completely leached. In all cases, the direction of the NRM does not change appreciably. Such a response suggests that much of the magnetization characteristic of Chuar Group strata is carried in detrital hematite.

5. Discussion

Two characteristic remanence magnetizations were identified in the Chuar Group (from the Galeros and Kwagunt Formations), which provide two distinct ancient field determinations for Laurentia during the Neoproterozoic. The older Galeros Formation is defined by 16 sites with a mean paleomagnetic pole located in the equatorial central Pacific (Table 2; Fig. 9). The Kwagunt Formation is defined by nine sites with a mean paleomagnetic pole also located in the central Pacific but with a significantly higher

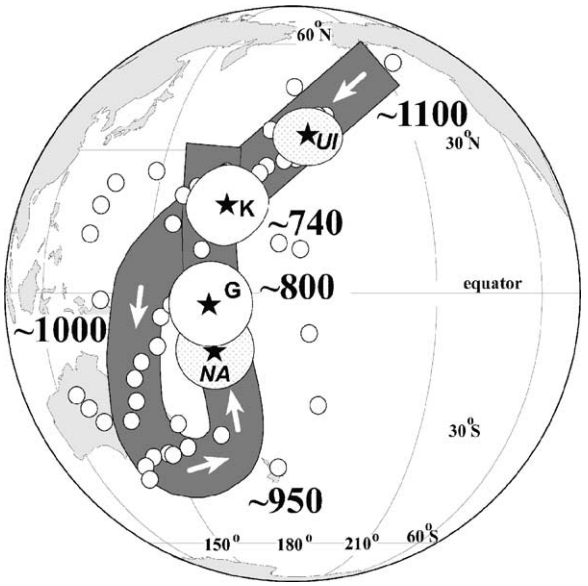


Fig. 9. Summary apparent polar wander path for Laurentia (orthographic projection) for the 1100–700 Ma time interval. Open circles are selected Keweenawan and Grenville "A poles" from Weil et al. (1998) and Warnock et al. (2000). K, Kwagunt Formation, GCSG (this study); G, Galeros Formation, GCSG (this study); NA, Nankoweap Formation, GCSG (Weil et al., 2003); UI, Unkar Group Igneous Suite, GCSG (Weil et al., 2003).

north latitude (Table 2; Fig. 9). We interpret both Chuar formation poles to be primary based on the presence of dual-polarity magnetizations, a positive reversal test, positive fold tests, and field evidence that indicates that the main folding of the Chuar Group occurred synchronous with its deposition (Timmons et al., 2001). It should be noted that the Chuar Group paleomagnetic poles are similar to Late Cambrian and Early Ordovician mean paleopoles for North America, but considerably different from latest Precambrian (~580 Ma) paleomagnetic poles (e.g., Van der Voo, 1993; Meert et al., 1994), which are located at much higher south latitudes. However, any remagnetization event with an age younger than Precambrian can be ruled out because the ancient magnetization in the Chuar Group strata is demonstrated to be pre-folding (and hence older than ~740 Ma), and Chuar strata lie in angular unconformity below the flat-lying Middle Cambrian Tapeats sandstone of the Tonto Group.

The similarity between Grand Canyon Supergroup poles (this study; Weil et al., 2003) and those poles from the Keweenawan rift system and Grenville province of northeast North America has long been noted (e.g., Elston, 1993). This similarity suggests that there has been at most only a modest magnitude of rotation of the Colorado Plateau with respect to the stable interior of North America during the Phanerozoic, despite the fact that some researchers still argue for up to 10–15° of clockwise rotation of the plateau based on paleomagnetic pole comparisons between Upper Triassic strata from the southwest and the east coast of North America (e.g., Kent and Witte, 1993; Kent and Olsen, 1997; Steiner and Lucas (2000), Steiner (2003)). Although the comparison between limited Proterozoic data from the Grand Canyon Supergroup with the track of poles defined by rocks from northeastern North America does not allow for a robust estimate of the magnitude of plateau rotation, based on the recent work of Molina Garza et al. (1998) we assume a negligible magnitude (less than 6°) of overall clockwise rotation of the Colorado Plateau since the time of Chuar Group deposition.

The upper age limit of the younger Chuar Group pole is ~740 Ma, based on a recent U–Pb zircon age determination from an ash bed in the Walcott member of the Kwagunt Formation (Karlstrom et al., 2000). The age of the lower part of the Galeros Formation (older Chuar Group pole) is estimated to be ~800 Ma (Karlstrom et al., 2000) based on the thickness of the Group, the lack of any observable internal unconformities, and average Phanerozoic shallow marine sedimentation rates (Sadler, 1981). However, with no geochronologic ages from the Galeros Formation a precise age for the formation is undeterminable. The lower age limit for the Galeros Formation is based on a 1090 Ma age from the upper Unkar Group (Weil et al., 2003), which is separated from the lower Chuar Group by the 113–250 m-thick Nankoweap Formation and an angular unconformity.

Preliminary paleomagnetic analysis of Chuar (and other Grand Canyon Supergroup) rocks was previously carried out by Elston (1989a) and Elston (1993), and represented an initial attempt at defining the Neoproterozoic APWP of Laurentia. However, many details of these results (i.e., demagnetization behavior, specific field-based tests for the antiquity of the magnetizations, identification of carriers of magnetizations, etc.) have never fully been published in peer-reviewed journals, or have only been published

in abstract form (Elston, 1986), in field guides (Elston, 1989b), or simply as a compilation of data, with little explanation (Elston, 1993; Elston et al., 2002). Comparison of results from this study with a recent table of poles published by Elston et al. (2002, their Table 4) reveals a significant difference in paleolatitude between the two datasets. The results compiled by Elston et al. (2002) show a consistent south-directed offset compared to the results presented here (about 16° for the Kwagunt Formation and about 10° for the Galeros Formation). Because only pole positions were reported it is impossible to determine the source of the discrepancy between the two datasets.

5.1. Laurentia's APWP

The time progression of paleopoles that define Laurentia's APWP for the 1000–700 Ma time interval has remained ambiguous and has generated considerable debate over the past few decades. Although most researchers agree that the Grenville section of Laurentia's APWP loops down (southward) through the southwest Pacific quadrant (Fig. 9), both the geometry and age progression of the path are controversial because of the high degree of age uncertainty for specific results (Weil et al., 1998; Walderhaug et al., 1999). This uncertainty is based in large part on the fact that most of the Grenville Province paleomagnetic data have come from metamorphic rocks, and therefore the timing of magnetization acquisition can be ambiguous (Warnock et al., 2000). The temporal ambiguity of paleomagnetic poles from the Grenville section of Laurentia's APWP has produced two disparate proposals regarding the path's geometry and direction of younging: (1) a clockwise loop (Fig. 10a) (McWilliams and Dunlop, 1978; Berger et al., 1979; Hyodo and Dunlop, 1993; Alvarez and Dunlop, 1998) and (2) a counterclockwise loop (Fig. 10b and c) (Weil et al., 1998; Warnock et al., 2000). These conflicting APWP have major implications and consequences for how paleomagnetic-based configurations for Rodinia are constructed. The number of well-dated robust paleomagnetic data for the relevant time period is small; using only key poles, it is nearly impossible to base reconstructions on the few results from a given continent for the purposes of paleogeography (Fig. 10d) (Buchan et al., 2001). However, if several high-quality poles are available for a given pair of continents, and a clearly defined APWP path can be generated for each block, then APWP geometry becomes a useful tool for generating relative continental reconstructions (Weil et al., 1998). Consequently, it is imperative that the correct time progression and geometry of Laurentia's APWP be known before further speculation is made on the paleogeography of the Rodinia supercontinent during its lifecycle and ultimately during its rift-to-drift history.

Sampling multiple horizons of Galeros and Kwagunt strata within the Chuar Group has provided two well-defined paleomagnetic poles for this critical interval of time during the Neoproterozoic, between about 800 and 740 Ma. Because of the sedimentary nature of the Chuar Group, the sequence of poles obtained, assuming a primary acquisition, provides a relative age progression for a part of Laurentia's APWP (Fig. 9). Utilizing recent paleomagnetic results from the southwest US (Weil et al., 2003) and other North American Proterozoic results (summarized in Weil et al., 1998), we argue that the most internally consistent explanation for the data is a tight counterclockwise APWP loop that passes down through the south Pacific (Fig. 9). It should be noted that this preferred counterclockwise path is specifically dependent on the ages of the poles chosen for the Laurentian APWP for the interval from 1000 to 950 Ma. These poles are from the Grenville metamorphic rocks in Ontario, for which ages have been interpolated from regional cooling ages (Cosca et al., 1991, 1992, 1995; Busch et al., 1996).

5.2. Paleogeography and Rodinia

The break-up history of Rodinia is not well understood due to the paucity of well-dated Neoproterozoic paleomagnetic poles. What we do know is that Laurentia and Baltica drifted into high southern latitudes by the latest Precambrian (Torsvik et al., 1996), and that the Laurentian Iapetus rift margin formed between Laurentia, West Gondwana and Baltica by ca. 580 Ma (Cawood et al., 2001). Furthermore, it is now accepted that the final assembly of Gondwana occurred at ca. 560 Ma with the collision of the West Gondwana cratons, the Arabian–Nubian Shield, Madagascar, Australia, Antarctica and India (e.g., Meert and Powell, 2001; Meert, 2001). Meert (2003) provides a thorough review of the events related to the assembly

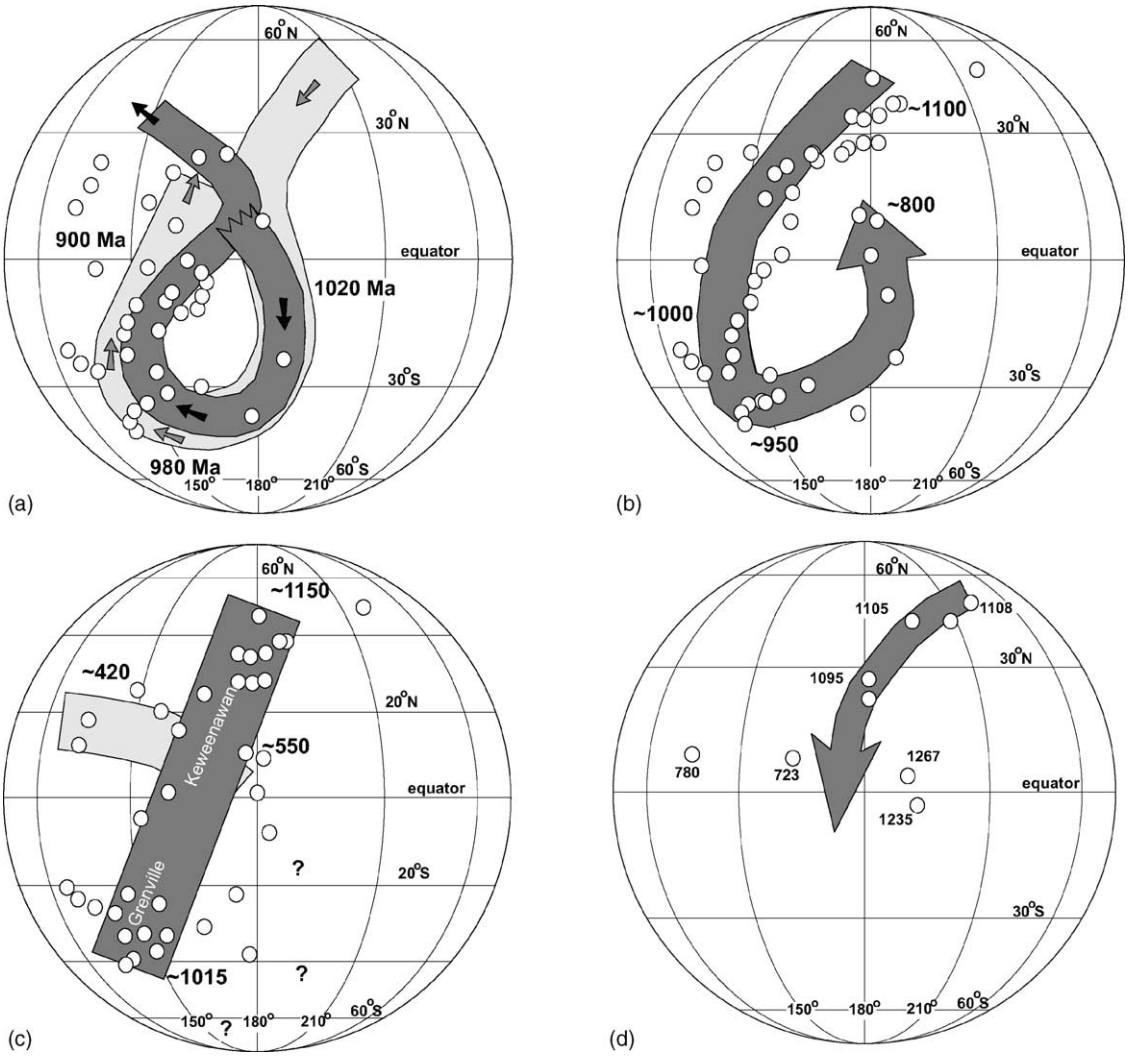


Fig. 10. Several previously proposed Proterozoic APWPs for Laurentia. (a) Clockwise loop (dark grey path) based on Grenville poles used by Hyodo and Dunlop (1993) and Costanzo and Dunlop (1998). Also shown is a similar clockwise APWP proposed by Roy (1983) (medium grey). (b) Counterclockwise loop for the 1100–800 Ma time interval based on similarities between Grenville and Keweenawan paleopoles from North America with similar aged poles from Baltica (Weil et al., 1998). (c) Keweenawan and Grenville "A poles" used by Warnock et al. (2000). (d) Key paleopoles for Laurentia for part of the 1200–500 Ma period as proposed by Buchan et al. (2001) (arrow represents well-defined Keweenawan track).

of Eastern Gondwana. We still do not know the exact timing of Rodinia's breakup, especially along the Cordilleran margin of Laurentia; what blocks actually rifted and drifted away and from where; and whether or not the rifted blocks behaved as a coherent, large continental mass (i.e., East Gondwana) or as a series of smaller, independent fragments (e.g., separate Australia, India, South China, and Seychelles blocks).

To date, high-quality paleomagnetic results have been determined for about 750 Ma for the Australian craton (Wingate and Giddings, 2000), the South China block (Evans et al., 2000), the India subcontinent (Torsvik et al., 2001a), and the Seychelles microcontinent (Torsvik et al., 2001b). However, there are no robust well-dated paleomagnetic data for this time period from Antarctica, Siberia, Madagascar, or any

of the West Gondwana blocks (Amazonia, Rio de la Plata, West Africa, Congo, Sao Francisco, or Kalahari cratons) (Torsvik et al., 2001a). The new results from the East Gondwana blocks (Australia, India, and the Seychelles blocks) and South China have spawned renewed debate as to the configuration of western (in North American coordinates) Rodinia prior to its breakup. Previously, the SWEAT (Southwest United States—East Antarctica) hypothesis, which placed the western margin of Laurentia against East Antarctica (Moores, 1991), had been the preferred reconstruction. Other researchers have recently argued for a more southerly position for the East Gondwana blocks (AUSWUS or Australia—Southwest United States), mainly Australia and East Antarctica, based on isotopic and geochronologic mapping (Borg and DePaolo, 1994), and the truncation of deformation and sedimentary belts along opposing ancient cratonic margins (Karlstrom et al., 1999, 2001; Burrett and Berry, 2000). More recently, Wingate et al. (2002), based on new paleomagnetic and geochronology results from Western Australia, proposed an even more southerly position of Australia/East Antarctica, against present-day Mexico (AUSMEX). The growing list of acronyms supports the contention that the details of the existence and makeup of Rodinia are

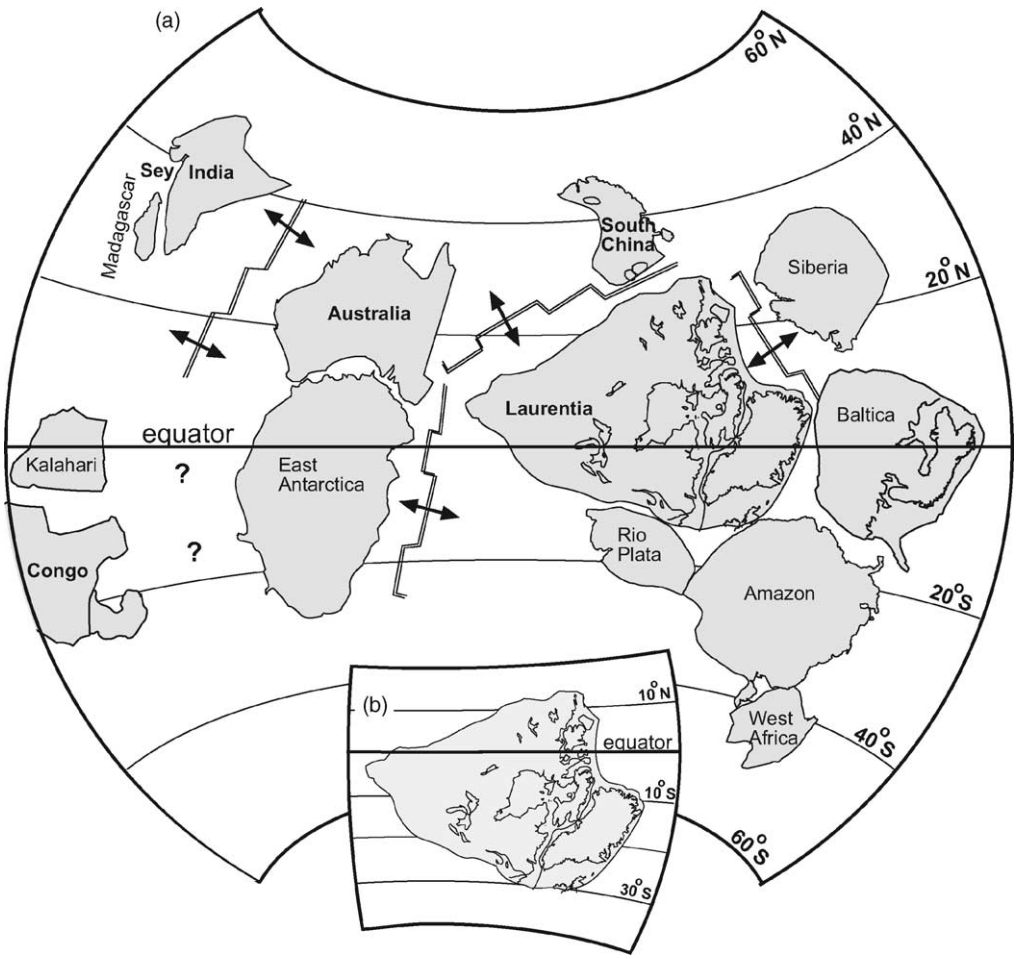


Fig. 11. (a) Reconstruction at ca. 750 Ma based on paleomagnetic data from Laurentia (this study; Kwagunt Formation), Australia (Wingate and Giddings, 2000), South China (Evans et al., 2000), and India/Seychelles (Torsvik et al., 2001a). (b) Paleolatitudinal position of Laurentia based on results from the Galeros Formation.

still unclear and in dire need of more well-dated paleomagnetic results.

In both the SWEAT and AUSWUS fits, the relative configuration of East Gondwana is the same, as it is treated as a coherent block. However, the new East Gondwana results require a drastically different reconstruction subsequent to rifting between Australia-East Antarctica and the India/Madagascar/Seychelles continent (Torsvik et al., 2001b; Torsvik, 2003). The data from Australia (Mundine Well Dike Swarm; Wingate and Giddings, 2000), India (Torsvik et al., 2001a), and the Seychelles microcontinent (Torsvik et al., 2001b) for ∼755 Ma are not consistent with either the SWEAT or the AUSWUS connection and indicate that Australia, and consequently whatever cratons were outboard of Australia, had already completed the rift-to-drift transition from Laurentia by ∼755 Ma.

When the new paleomagnetic results from the Chuar Group are compared with recent results from several of the East Gondwana blocks and South China it is clear that regardless of the preferred Rodinia reconstruction (e.g., SWEAT, AUSWUS, AUSMEX, etc.) the present-day Cordilleran margin of Laurentia had already undergone its rift-to-drift transition from an opposing margin well before 750 Ma, because in a 750 Ma reconstruction (Fig. 11) based on available well-dated paleomagnetic poles these continents are far apart. Presently, a dearth of paleomagnetic data for the major constituents of Rodinia at about 800 Ma prevents any robust reconstruction for the Rodinia supercontinent prior to its breakup.

6. Conclusions

Paleomagnetic results from the Chuar Group, Grand Canyon Supergroup, yield two primary paleomagnetic poles for Laurentia for the Neoproterozoic. The pole positions for the two Chuar Group formations, in combination with other recent Late Proterozoic paleomagnetic results, define a counterclockwise loop in the apparent polar wander path for North America in the south Pacific (Fig. 9). This is in contrast to recently published clockwise APWP loops for Laurentia for the 1100–700 Ma time interval. Furthermore, our data are of about the same age as recently published data from East Gondwana, which together suggest that drifting had begun between Laurentia and the East Gondwana block(s) well before 750 Ma. As a whole, the Rodinia Supercontinent drifted northward from high southern latitudes to equatorial latitudes during the 950–750 Ma time interval. This drift history positioned the supercontinent at or near the equator between 800 and 740 Ma, which does not contradict the hypothesized Sturtian aged snowball Earth event, the first of several widespread glaciations in the Neoproterozoic (Hoffman et al., 1998). At the same time, extension initiated along the Cordilleran margin of Laurentia, represented by the deposition of the Chuar Group and possibly the Pahrump and Uinta Mountain groups of the western U.S. This large-scale extensional event culminated in the dispersal of Laurentia away from the East Gondwana blocks before ∼750 Ma.

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References

Alvarez, V.C., Dunlop, D.J., 1998. A regional paleomagnetic study of lithotectonic domains in the Central Gneiss Belt, Grenville Province. Ontario. Earth Planet. Sci. Lett. 157, 103.

Bell, R.T., Jefferson, C.W., 1987. A hypothesis for an Australian–Canadian connection in the late Proterozoic and the birth of the Pacific Ocean. In: Edward, B. (Convener), Proceedings of the Pacific Rim Congress'87: An International Congress on the Geology, Structure, Materialization and Economics of the Pacific Rim. pp. 39–50.

Berger, G.W., York, D., Dunlop, D.J., 1979. Calibration of Grenvillian paleopoles by $^{40}Ar/^{39}Ar$ dating. Nature 277, 46–48.

Bond, G.C., Kominz, M.A., 1984. Construction of tectonic subsidence curves for the early Paleozoic miogeocline, southern

Canadian Rocky Mountains: implications for subsidence mechanisms, age of break-up, and crustal thinning. Geol. Soc. Am. Bull. 95, 155–173.

Bond, G.C., Nickeson, P.A., Kominz, M.A., 1984. Breakup of a supercontinent between 625 and 555 Ma: new evidence and implications for continental histories. Earth Planet. Sci. Lett. 70, 325–345.

Borg, S.G., DePaolo, D.J., 1994. Laurentia, Australia, and Antarctica as a Late Proterozoic supercontinent: constraints from isotopic mapping. Geology 22, 307–310.

Buchan, K.L., Ernst, R.E., Hamilton, M.A., Mertanen, S., Pesonen, L.J., Elming, S., 2001. Rodinia: the evidence from integrated palaeomagnetism and U–Pb geochronology. Precambrian Res. 110, 9–32.

Burrett, C., Berry, R., 2000. Proterozoic Australia–western US (AUSWUS) fit between Laurentia and Australia. Geology 28, 103–106.

Busch, J.P., van der Pluijm, B.A., Hall, C.M., Essene, E.J.L., 1996. Normal faulting revealed by $^{40}Ar/^{39}Ar$ thermochronology near the Robertson Lake shear zone, Grenville orogen, Canada. Tectonics 15, 387–402.

Cawood, P.A., McCausland, P.J.A., Dunning, G.R., 2001. Opening Iapetus: constraints from the Laurentian margin in Newfoundland. Geol. Soc. Am. Bull. 113, 443–452.

Cosca, M.A., Sutter, J.F., Essene, E.J., 1991. Cooling and inferred uplift/erosion history of the Grenville orogen Ontario: constraints from $^{40}Ar/^{39}Ar$ thermochronology. Tectonics 10, 959–977.

Cosca, M.A., Essene, E.J., Kunk, M.J., Sutter, J.F., 1992. Differential unroofing within the Central Metasedimentary belt of the Grenville orogen: constraints form $^{40}Ar/^{39}Ar$ thermochronology. Contrib. Mineral Petrolol. 110, 211–225.

Cosca, M.A., Essene, E.J., Mezger, K., van der Pluijm, B.A., 1995. Constraints on the duration of tectonic processes: protracted extension and deep-crustal rotation in the Grenville orogen. Geology 23, 361–364.

Costanzo, A.V., Dunlop, D.J., 1998. A regional paleomagnetic study of lithotectonic domains in the Central Gneiss Belt, Grenville Province, Ontario. Earth Planet. Sci. Lett. 157, 89–103.

Cook, D.A., 1991. Sedimentary and shale petrology of the upper Proterozoic Walcott Member, Kwagunt Formation, Chuar Group, Grand Canyon, Arizona. Masters Thesis, Flagstaff, Northern Arizona University, Arizona, 128 pp.

Dalziel, I.W.D., 1997. Neoproterozoic–Paleozoic paleogeography and tectonics: review, hypothesis, and environmental speculation. Geol. Soc. Am. Bull. 109, 16–42.

Dalziel, I.W.D., 1991. Pacific margins of Laurentia and East Antarctica–Australia as a conjugate rift pair: evidence and implications for an Eocambrian supercontinent. Geology 19, 598–601.

Dehler, C.M., Des Marais, D.J., Bowring, S., Sharp, Z., Kalrstrom, K.E., Elrick, M., 1999. Chuar Group (1.1–0.74) Ga, Grand Canyon: carbon-isotope systematics at the onset of Sturtian glaciation. Geol. Soc. Am. Abst. Prog. 31, A-487.

Dehler, C.M., Elrick, M., Karlstrom, K.E., Smith, G.A., Crossey, L.J., Timmons, J.M., 2001a. Neoproterozoic Chuar Group (~800–742 Ma), Grand Canyon: a record of cyclic marine deposition during global cooling and supercontinent rifting. Sedimentary Geol. 141–142, 465–499.

Dehler, C.M., Prave, A., Crossey, L.J., Karlstrom, K.E., 2001b. Linking Mid-Neoproterozoic successions in the western United States: The Chuar Group–Uinta Mountain Group–Pahrump Group connection (CHUMP). Geol. Soc. Am. Abstr. Prog. 33, A20–A21.

Elston, D.P., 1986. Magnetostratigraphy of Late Proterozoic Chuar Group and Sixtymile Formation, Grand Canyon Supergroup, northern Arizona; correlation with other Proterozoic strata of North America. Geol. Soc. Am. Abstr. Prog. 18 (5), 353.

Elston, D.P., 1989a. Grand Canyon Supergroup, northern Arizona; stratigraphic summary and preliminary paleomagnetic correlations with parts of other North American Proterozoic successions. In: Jenney, J.P., Reynolds, S.J. (Eds.), Geologic Evolution of Arizona, Arizona Geological Society Digest, vol. 17. pp. 259–272.

Elston, D.P., 1989b. Middle and Late Proterozoic Grand Canyon Supergroup. In: Elston, D.P., Billingsley, G.H., Young, R.A. (Eds.), Proceedings of the 28th International Geological Congress, Field Trip Guidebook T115/315: Geology of Grand Canyon, Northern Arizona. American Geophysical Union, Washington, DC, Chapter 9, pp. 94–105.

Elston, D.P., 1993. Middle and Early–Late Proterozoic Grand Canyon Supergroup, northern Arizona. In: Reed, J.C., Bickford, M.E., Houston, R.S., Link, P.K., Rankin, D.W., Sims, P.K., Van Schumus, W.R. (Eds.), Precambrian, Conterminous U.S.: Boulder, Colorado, Geological Society of America, Geology of North America, vol. C-2. pp. 521–529.

Elston, D.P., McKee, E.H., 1982. Age and correlation of the Proterozoic Grand Canyon disturbance, northern Arizona. Geol. Soc. Am. Bull. 93, 681–699.

Elston, D.P., Enkin, R.J., Baker, J., Kisilevsky, D.K., 2002. Tightening the belt: paleomagnetic–stratigraphic constraints on deposition, correlation, and deformation of the Middle Proterozoic (ca. 1.4 Ga) Belt-Purcell Supergroup, United States and Canada. Geol. Soc. Am. Bull. 114 (5), 619–638.

Evans, D.A.D., Li, Z.X., Kirschvink, J.L., Wingate, M.T.D., 2000. A high-quality mid-Neoproterozoic paleomagnetic pole from South China, with implications for ice ages and the breakup configuration of Rodinia. Precambrian Res. 100, 313–334.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. London A 217, 295–305.

Ford, T.D., Breed, W.J., 1973. Late Precambrian Chuar Group, Grand Canyon, Arizona. Geol. Soc. Am. Bull. 84, 1243–1260.

Halls, H.C., Pesonen, L.J., 1982. Paleomagnetism of Keweenawan rocks. Geol. Soc. Am. Memoir 156, 173–201.

Harland, W.B., 1964. Evidence of late Precambrian glaciation and its significance. In: Nairn, A.E.M. (Eds.), Problems in Palaeoclimatology. pp. 119–149.

Hartz, E.H., Torsvik, T.H., 2002. Baltica upside down: a new plate tectonic model for Rodinia and the Iapetus Ocean. Geology 30, 255–258.

Henry, S., 1979. Chemical demagnetization-methods, procedures, and applications through vector analysis. Can. J. Earth Sci. 16 (9), 1832–1841.

Hoffman, P.F., 1991. Did the breakout of Laurentia turn Gondwana inside out? Science 252, 1409–1412.

Hoffman, P.F., Kaufman, A.J., Halverson, G.P., Schrag, D.P., 1998. A Neoproterozoic snowball Earth. Science 281, 1342–1346.

Hoffman, P.F., 1999. The breakup of Rodinia, birth of Gondwana, true polar wander, and the snowball Earth. J. Afr. Earth Sci. 28, 9–26.

Hoffman, P.F., Schrag, D.P., 2002. The snowball Earth hypothesis: testing the limits of global change. Terra Nova 14, 129–155.

Hyodo, H., Dunlop, D.J., 1993. Effect of anisotropy on the paleomagnetic contact test for a Grenville dike. J. Geophys. Res. 98 (B5), 7997–8017.

Karlstrom, K.E., Williams, M.L., McLelland, J., Geissman, J.W., Ahall, K., 1999. Refining Rodinia: geologic evidence for the Australia–western U.S. connection in the Proterozoic. GSA Today 9, 1–7.

Karlstrom, E., Bowring, S.A., Dehler, C.M., Knoll, A.H., Porter, S.M., Sharp, Z., Des Marais, D.J., Weil, A.B., Geissman, J.W., Elrick, M., Timmons, M.J., Keefe, K., Crossey, L.J., 2000. The Chuar Group of the Grand Canyon: Record of break up of Rodinia, associated change in the global carbon cycle, and eukaryotic diversification by 740 Ma. Geology 28, 619–622.

Karlstrom, K.E., Ahall, K.-I., Harlan, S.S., Williams, M.L., McLelland, J., Geissman, J.W., 2001. Long-lived (1.8–0.8 Ga) Cordilleran type orogen in southern Laurentia, its extensions to Australia and Baltica, and implications for refining Rodinia. Precambrian Res. 111, 5–30.

Kent, D.V., Olsen, P.E., 1997. Paleomagnetism of continental sediments of Late Triassic age from Dan River-Danville Rift Basin (eastern North America). Geol. Soc. Am. Bull. 109, 366–379.

Kent, D.V., Witte, W.K., 1993. Slow apparent polar wander for North America in the Late Triassic and large Colorado plateau rotation. Tectonics 12, 291–300.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astronom. Soc. 62, 699–718.

Kirschvink, J.L., 1992. Late Proterozoic low-latitude global glaciation: the Snowball Earth. In: Schopf, J.W., Klein, C. (Eds.), The Proterozoic Biosphere. Cambridge University Press, Cambridge, pp. 51–52.

Li, Z.X., Zhang, L., Powell, C.M., 1996. Positions of the East Asian cratons in the Neoproterozoic supercontinent Rodinia. In: Li, Z.X., Metcalfe, I. Powell, C.M. (Eds.), Breakup of Rodinia and Gondwanaland and assembly of Asia; Aust. J. Earth Sci. 43, 593–604.

Lindsay, J.F., Korsch, R.J., Wilford, J.R., 1987. Timing the breakup of a Proterozoic supercontinent; evidence from Australian intracratonic basins. Geology 15 (11), 1061–1064.

Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties. Geophys. Res. Lett. 17, 159–162.

McElhinny, M.W., 1964. Statistical significance of the fold test in paleomagnetism. Geophys. J. R. Astronom. Soc. 8, 338–340.

McFadden, P.L., Lowes, F.J., 1981. The discrimination of mean directions drawn from Fisher distributions. Geophys. J. R. Astronom. Soc. 67, 19–33.

McWilliams, M.O., Dunlop, D.J., 1978. Grenville paleomagnetism and tectonics. Can. J. Earth Sci. 15, 687–695.

Meert, J.G., 2001. Growing Gondwana and rethinking Rodinia: a paleomagnetic perspective. Gondwana Res. 4, 279–288.

Meert, J.G., 2003. A synopsis of events related to the assembly of eastern Gondwana. Tectonophysics 362, 1–40.

Meert, J.G., Van der Voo, R., Payne, T.W., 1994. Paleomagnetism of the Catoctin volcanic province a new Vendian–Cambrian apparent polar wander path for North America. J. Geophys. Res. B (99), 4625–4641.

Meert, J.G., Van der Voo, R., Ayub, S., 1995. Paleomagnetic investigation of the Neoproterozoic Gagwe lavas and Mbozi complex, Tanzania and the assembly of Gondwana. Precambrian Res. 74, 225–244.

Meert, J.G., Van der Voo, R., 1996. Paleomagnetic and $^{40}Ar/^{39}Ar$ study of the Sinyai Dolerite, Kenya; implications for Gondwana assembly. J. Geol. 104, 131–142.

Meert, J.G., Powell, C.M., 2001. Editorial: assembly and break-up of Rodinia: introduction to the special volume. Precambrian Res. 110, 1–8.

Molina Garza, R.S., Acton, G.D., Grissman, J.W., 1998. Carboniferous through Jurassic paleomagnetic data and their bearing on rotation of the Colorado plateau. J. Geophys. Res. 103, 24179–24188.

Moores, E.M., 1991. Southwest U.S.–East Antarctica (SWEAT) connection: a hypothesis. Geology 19, 425–428.

Park, J.K., Aitken, J.D., 1986. Paleomagnetism of the Late Proterozoic Tsezotene Formation of Northwestern Canada. J. Geophys. Res. Solid Earth Planets 91 (B5), 4955–4970.

Piper, J.D.A., 1982. The Precambrian paleomagnetic record: the case for the Proterozoic supercontinent. Earth Planet. Sci. Lett. 59, 61–89.

Porter, S., Knoll, A.H., 2000. Testate amoebae in the Neoproterozoic Era: evidence from vase-shaped microfossils in the Chuar Group, Grand Canyon. Paleobiology 26, 345–370.

Powell, C.M., Li, Z.X., McElhinny, M.W., Meert, J.G., Park, J.K., 1993. Paleomagnetic constraints on timing of the Neoproterozoic breakup of Rodinia and the Cambrian formation of Gondwana. Geology 21, 889–892.

Powell, C., Preiss, W.V., Gatehouse, C.G., Krapez, B., Li, Z.X., 1994. South Australian record of a Rodinian epicontinental basin and its mid-Neoproterozoic breakup (~700 Ma) to form the Palaeo-Pacific Ocean. Tectonophysics 237, 113–140.

Reynolds, M.W., Palaces, J.G., Elston, D.P., 1988. Potential petroleum source rocks in the late Proterozoic Chuar Group Grand Canyon, Arizona. USGS Circular 1025, 49–50.

Roy, J.L., 1983. Paleomagnetism of the North American Precambrian: a look at the data base. Precambrian Res. 19, 319–348.

Sadler, P.M., 1981. Sediment accumulation rates and the completeness of stratigraphic sections. J. Geol. 89, 569–584.

Sears, J.W., 1990. Geologic structure of the Grand Canyon Supergroup. In: Beus, S.S., Morales, M. (Eds.), Grand Canyon Geology. Oxford University Press, New York, pp. 71–82.

Steiner, M.B., 2003. A Cratonic Middle Jurassic Paleopole: Callovian-Oxfordian stillstand (J-2 Cusp), Rotation of the Colorado Plateau, and Jurassic North American Apparent Polar Wander: Tectonics, vol. 22. p. 10.1029/2001TC001284.

Steiner, M.B., Lucas, S.G., 2000. Paleomagnetism of the Late Triassic Petrified Forest Formation, Chinle Group, western

United States: further evidence of “large” rotation of the Colorado Plateau. J. Geophys. Res. 105, 25791–25808.

Stern, R.J., 1994. Arc-assembly and continental collision in the Neoproterozoic East African Orogen, implications for the consolidation of Gondwanaland. Annu. Rev. Earth Planetar. Sci. 22, 319–351.

Tauxe, L., 1998. Paleomagnetic Principles and Practice, Modern Approaches in Geophysics, vol. 17. Kluwer Academic Publishers, London, 299 pp.

Timmons, J.M., Karlstrom, K.E., Dehler, C.M., Geissman, J.W., Heizler, M.T., 2001. Proterozoic multistage (ca. 1.1 and 0.8 Ga) extension recorded in the Grand Canyon Supergroup and establishment of northwest- and north-trending tectonic grains in the southwestern United States. Geol. Soc. Am. Bull. 113, 163–1800.

Torsvik, T.H., Smethurst, M.A., Meert, J.G., Van der Voo, R., McKerrow, W.S., Braiser, M.D., Sturt, B.A., Walderhaug, H.J., 1996. Continental break-up in the Neoproterozoic and Paleozoic—a tale of Baltica and Laurentia. Earth-Sci. Rev. 40, 229–258.

Torsvik, T.H., Carter, L.M., Ashwal, L.D., Bhushan, S.K., Pandit, M.K., Jamtveit, B., 2001a. Rodinia refined or obscured: palaeomagnetism of the Malani igneous suite (NW India). Precambrian Res. 108, 319–333.

Torsvik, T.H., Ashwal, L.D., Tucker, R.D., Eide, E.A., 2001b. Neoproterozoic geochronology and palaeogeography of the Seychelles microcontinent: the India link. Precambrian Res. 110, 47–59.

Torsvik, T.H., 2003. The Rodinia jigsaw puzzle. Science 300, 1379–1381.

Van der Voo, R., 1993. Paleomagnetism of the Atlantic, Tethys and Iapetus Oceans. Cambridge University Press, New York, 411 pp.

Vidal, G., Knoll, A.H., 1983. Proterozoic plankton. Geol. Soc. Am. Memoir 161, 265–277.

Vidal, G., Ford, T.D., 1985. Microbiotas from the Late Proterozoic Chuar Group (northern Arizona) and Uinta Mountain Group (Utah) and their chronostratigraphic implications. Precambrian Res. 28, 349–389.

Walderhaug, H.J., Torsvik, T.H., Eide, A.E., Sundvoll, B., Binger, B., 1999. Geochronology and palaeomagnetism of the Hunnedalen dykes, SW Norway: implications for the Sveconorwegian apparent polar wander loop. Earth Planet. Sci. Lett. 169, 71–83.

Warnock, A.C., Kodama, K.P., Zeitler, P.K., 2000. Using thermochronology and low-temperature demagnetization to accurately date Precambrian poles. J. Geophys. Res. 105 (B8), 19,435–19,453.

Weil, A.B., Van der Voo, R., Mac Niocaill, C., Meert, G.M., 1998. The Proterozoic supercontinent Rodinia: Paleomagnetically derived reconstructions for the 1,100 to 800 Ma interval. Earth Planet. Sci. Lett. 154, 13–24.

Weil, A.B, Geissman, J., Heizler, M., Van der Voo, R.A., 2003. Paleomagnetic investigation of Middle Proterozoic mafic intrusions and Upper Proterozoic redbeds from the Lower Grand Canyon Supergroup, Arizona. Tectonophysics 375, 199–220.

Wingate, M.T.D., Giddings, J.W., 2000. Age and paleomagnetism of the Mundine Well dyke swarm, Western Australia: implications for an Australia-Laurentia connection at 755 Ma. Precambrian Res. 335–357.

Wingate, M.T.D., Pisarevsky, S.A., Evans, A.D., 2002. Rodinia connections between Australia and Laurentia: no SWEAT, no AUSWUS? Terra Nova 14 (2), 121–128.

Zijderveld, J.D.A., 1967. AC demagnetization of rocks: analysis of results. In: Collison, D.W., Runcorn, S.K., Creer, K.M. (Eds.), Methods in Paleomagnetism. Elsevier, New York,pp. 254–286.